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The effect of repeated cold and hot water immersion after muscle damage in women: a randomised controlled trial.

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Purpose

Exercise-induced muscle damage (EIMD) can result from unaccustomed or high-intensity eccentric exercise and is typically evaluated using objective and subjective markers of recovery. Cold and hot water immersion (CWI and HWI, respectively) have emerged as potential post-exercise recovery interventions to accelerate recovery between training sessions or competitions, but evidence is lacking, particularly in the female population. Therefore, this study aimed to compare the effect of CWI and HWI in enhancing recovery after EIMD compared to a passive control group (CON) in women.

Methods

Thirty women (23.4±2.9 years), randomly assigned to CWI, HWI or CON completed a standardised EIMD protocol. Immediately and 120 min after the EIMD, participants underwent a 10 min recovery intervention (CWI, HWI, or CON). Acute physiological response and recovery measures were assessed during a 72-hour follow-up period.

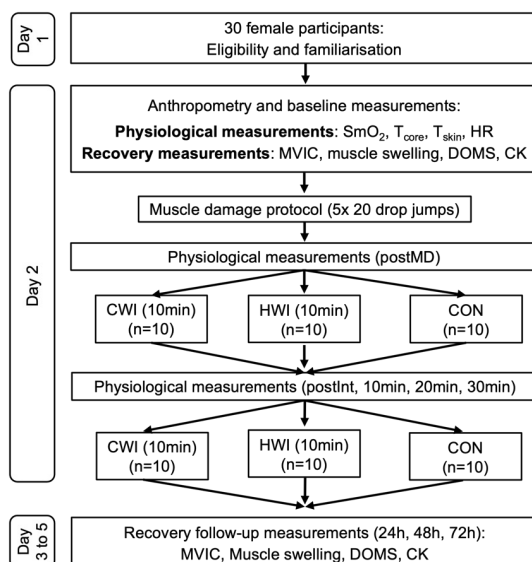


Fig. 1. Schematic illustration of the experimental protocol.

Results

SmO₂ was significantly lower in CWI compared to HWI (20 min: 67.8±2.7% vs. 74.7±3.2%, p=0.006; 30 min: 64.4±6.4% vs. 74.4±3.9%, p=0.001) and compared to CON (20 min: 73.5±6.6%, p=0.026; 30min: 73.0±6.1%, p=0.006). Core temperature was significantly higher in HWI compared to CWI at postInt and 30min and compared to CON at

postInt and throughout the 30 min follow-up. No difference between CWI and CON was detected for core temperature. Skin temperature was significantly reduced in CWI compared to HWI and CON between postInt and 30 min follow-up (all p<0.05). No significant differences were observed between CWI and HWI in objective and subjective recovery markers throughout the 72-hour follow-up.

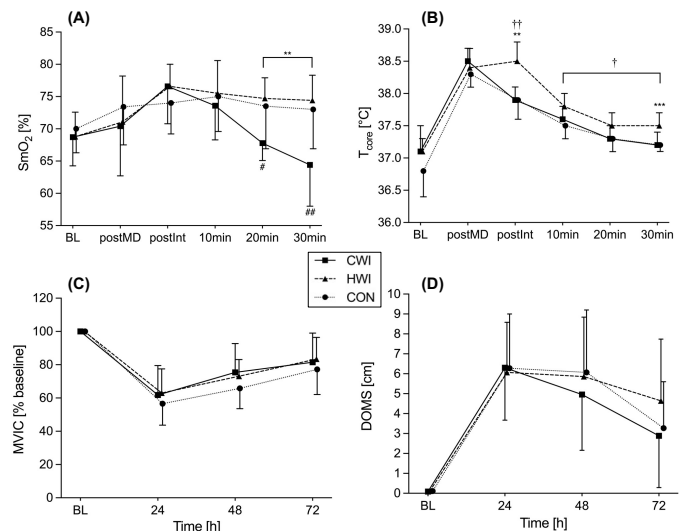


Fig. 2. Results (Mean ± SD) of (A) muscle oxygen saturation (SmO₂) (B) core temperature (T_{core}), (C) maximal voluntary isometric contraction (MVIC), and (D) delayed onset muscle soreness (DOMS) in function of time. BL baseline, postMD post muscle damage protocol, postInt post-intervention. Significant between-group differences: ***p<0.001 CWI vs HWI, **p<0.01 CWI vs HWI, ###p<0.01 CWI vs CON, #p<0.05 CWI vs CON, †p<0.01 HWI vs CON, *p<0.05 HWI vs CON.

Findings / Conclusion

- Following EIMD acute physiological responses in SmO₂, core- and skin temperature significantly differ between CWI and HWI
- Despite these physiological differences, our research found that neither CWI nor HWI demonstrated superior recovery benefits over CON during the 72-hour follow-up

Based on our findings, we cannot recommend the use of CWI or HWI to accelerate recovery after EIMD. However, psychological aspects associated with the immersion temperature might influence recovery, which should be considered.

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